Reading the genomic signatures of natural selection with machine learning

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Natural selection leaves distinctive signatures in genomes, but identifying these signatures and determining which evolutionary processes produced them remain challenging because of demographic history and other sources of genomic variation. In this talk, I will describe three machine learning approaches that address different challenges in detecting and interpreting these evolutionary footprints. I will first introduce TrIdent, a transfer learning approach that efficiently detects genomic regions shaped by natural selection from patterns of haplotype variation. I will then discuss PULSe, which uses positive-unlabeled learning to identify selective sweeps without requiring an explicitly defined neutral background. Finally, I will present SHADES, which uses transfer learning on representations of genomic summary statistics designed to distinguish among neutrality, balancing selection, selective sweeps, and adaptive introgression. Together, these approaches illustrate how machine learning can move beyond detecting genomic regions influenced by selection toward identifying the evolutionary processes that shaped them.